

PORTFOLIO

Marketing Automation & Integration Workflows

Five production-grade automations and a custom Marketo email-build skill, across marketing operations, lead-lifecycle orchestration and AI-assisted data enrichment.

PLATFORMS Workato · n8n

DOMAINS Marketo · Google Sheets · Gemini / LLM

GENERATED June 2026

All company names, credentials, tokens, field names and identifiers have been replaced with neutral placeholders (Company A, Company B, <REDACTED_API_TOKEN>, ...). Nothing confidential is included.

Overview

#	Workflow	Platform	What it solves
1	Source → Destination Marketo Sync	Workato	Hand leads between two Marketo instances and route them into the right Smart Campaign
2	Carrier / Shipper / Junk Qualification Agent	n8n	Classify a lead's domain as carrier, shipper or junk, enrich it, and write the result straight back to Marketo
3	Email Send-Time Optimization	n8n	Compute the single best hour to e-mail a contact
4	Domain Checker Skill (Arc agent ↔ Claude)	n8n · Arc	Expose the classifier as an Arc skill; Claude calls it and turns the returned JSON into a report
5	Email TLD → Country Updater	n8n	Derive a lead's country from its email ccTLD and write it back to Marketo
6	Marketo Email Builder (Word brief → Marketo)	Claude skill · Marketo MCP	Turn a Word brief into a Marketo-ready email against a master template, in one or many languages

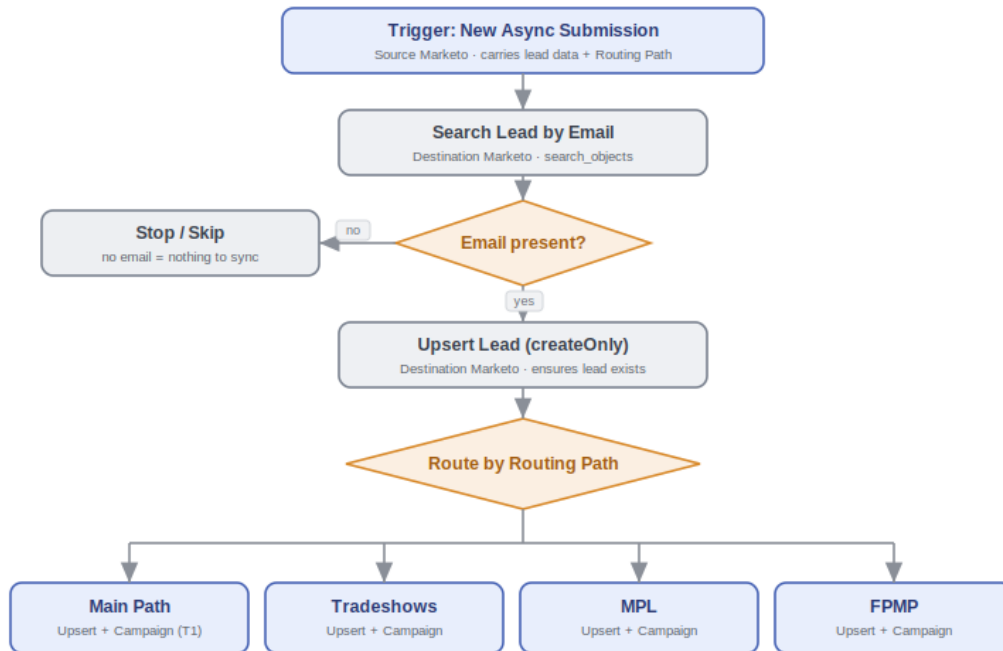
Skills demonstrated

- **Integration platforms:** Workato recipes & connections with data-pill mapping and concurrency control; n8n webhooks, Set/IF nodes, HTTP Request, Google Sheets and (where useful) LangChain agent nodes.
- **Marketing automation:** Marketo Smart Campaign triggering, upsert strategies (createOnly / updateOnly / createOrUpdate), cross-instance field mapping, OAuth2 token handling and webhook round-trips.
- **AI / LLM engineering:** web-search-grounded classification with brand-name fallback, three-tier deterministic-then-AI routing, strictly constrained structured outputs, and exposing a workflow as a synchronous tool an AI agent can call.
- **Data & ops patterns:** deterministic-first / AI-fallback logic, self-learning write-back that grows a reference dataset, idempotent record handling, and resilient multi-branch merges.

Workflow 1 — Source Marketo → Destination Marketo Sync

Platform: Workato (low-code iPaaS) · Recipe version: 131 · Concurrency: 10

Workflow 1 — Source Marketo → Destination Marketo Sync (Workato)



What it does

Two business units each run their **own, separate Marketo instance**. When a lead does something meaningful in **Company A's** Marketo (the source), this recipe hands that lead off in real time to **Company B's** Marketo (the destination), ensures the person record exists there, copies the relevant attributes across, and drops them into the correct **Smart Campaign** based on a routing key. In short: a **cross-instance lead handoff with conditional routing**.

Step-by-step logic

1. **Search lead by e-mail** in Destination Marketo (**search_objects** on the **leads** object, filtered by **email**), establishing whether the person already exists.
2. **Guard — “Email present?”** If the inbound record has no e-mail there is nothing to sync, so the recipe stops.
3. **Ensure the lead exists** — an upsert with **operation_type = createOnly** (lookup on **email**) creates the person only if new, mapping core identity fields.
4. **Route by TP Routing Path** — a chain of conditional branches, each performing a targeted update upsert and triggering a Smart Campaign:

Routing Path	Upsert mode	Notable field mapping	Then
Main Path	updateOnly	last-known product interest, person notes	Trigger campaign (ID: ••••) — Products Interest (T1), then stop
Tradeshows	updateOnly	person notes (comment capture), last-known campaign	Trigger campaign (ID: ••••), then stop
MPL	updateOnly	leadScore = 100, aqlstatus = known, last-known campaign	Trigger campaign (ID: ••••), then stop
FPMP	createOrUpdate	last-known campaign = Freight Marketplace Contact Request, source channel, product interest	Trigger campaign (ID: ••••), then stop

Each branch ends with an explicit **stop** so a lead only ever travels one path.

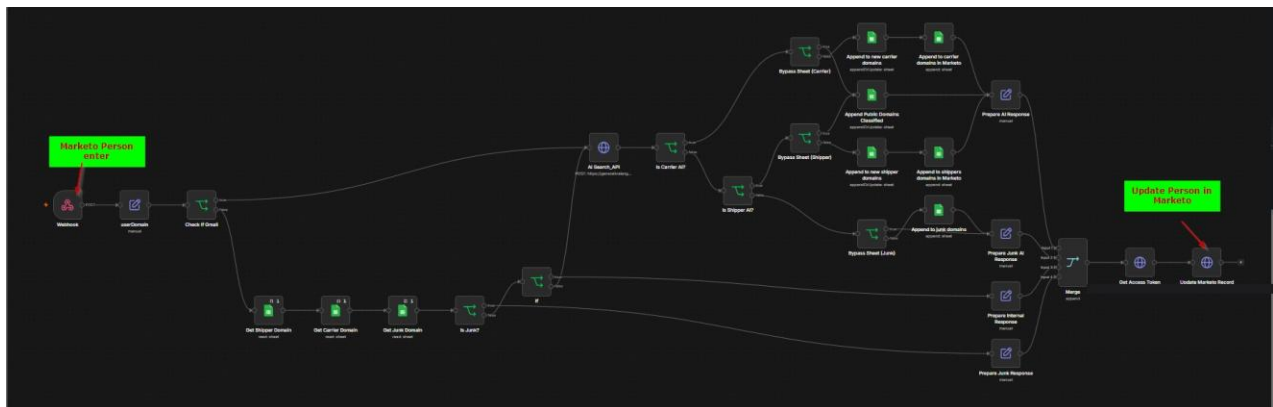
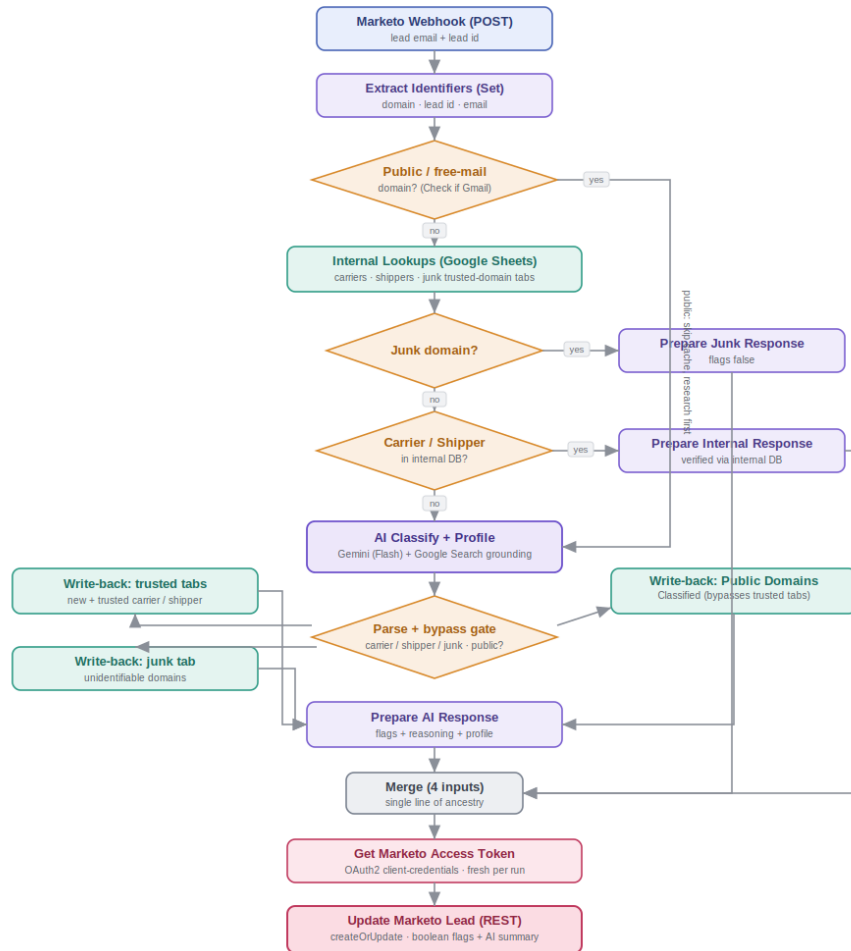
Why it's interesting

It solves a genuinely awkward problem — two Marketo orgs that need to behave like one — without custom code, while keeping the branching logic auditable and record operations safe to retry.

Workflow 2 — Carrier / Shipper / Junk Qualification Agent

Platform: n8n · **Trigger:** Marketo webhook (POST) · **Pattern:** three-tier lookup → AI fallback → closed-loop Marketo write-back

Workflow 2 — Carrier / Shipper / Junk Qualification Agent (n8n)



The production Marketo → n8n → Marketo canvas: a person enters from Marketo, is classified through the carrier / shipper / junk and public-domain tiers, and the record is updated back in Marketo (host redacted).

What it does

In logistics it matters enormously whether an inbound lead is a **carrier** (a company that moves freight) or a **shipper** (a company that makes or sells goods and needs them moved) — and whether the domain is **junk** (a free-mail, parked or otherwise unusable domain). Marketo calls this workflow with a lead's e-mail and id; the agent isolates the company domain, classifies it, builds a structured corporate profile, and **writes the verdict straight back onto the Marketo lead record** — using fast, free internal lookups first and only paying for an AI call when the domain is genuinely unknown.

This is the production-hardened evolution of the original classifier: it adds a dedicated **junk tier**, **public-domain routing** (free-mail domains are researched but kept out of the trusted-domain cache), a **self-learning write-back** into multiple reference tabs, and a **direct REST round-trip into Marketo** (OAuth token retrieval + lead update) instead of merely answering the webhook.

Step-by-step logic

1. **Marketo webhook (POST)** receives the lead e-mail and lead id in the request body.
2. **Extract identifiers (Set)** — parses the @domain, lead id and email from the payload, tolerating both structured and flattened body shapes.
3. **Public / free-mail check (Check if Gmail)**. The domain is tested against a list of public / free-mail providers. A **public domain** (e.g. a generic webmail address) can never be a trusted company domain, so it skips the internal cache and goes straight to AI research to see whether the underlying company is still identifiable. Corporate domains take the fast internal-lookup path first.
4. **Internal lookups (Google Sheets)** — three sequential reads against the trusted-domain tabs: carriers, shippers and a junk-exclusion list, all matched on the domain.
5. **Junk check**. If the domain is on the junk list → Prepare Junk Response (flags false, source = internal junk database). Otherwise continue.
6. **Internal carrier / shipper check**. If the domain is found in the carrier or shipper tab → Prepare Internal Response (set `is_carrier` / `is_shipper`, status = “verified via internal database”). Fast, deterministic, free.
7. **AI classify + profile (fallback)**. Unknown domains hit an HTTP request to **Gemini with the Google Search tool enabled**, so the model researches the live company before answering. The prompt enforces a strict contract: the response must begin with `Carrier -`, `Shipper -` or `Junk -`, followed by reasoning (≤ 50 words) and a structured profile (company_name, industry_description, 4-digit NACE code + description, logistics_classification, entity_location, entity_annual_revenue_eur, ultimate_parent_name, ultimate_parent_global_revenue_eur). A brand-name fallback instructs the model to re-search the core brand if the raw domain resolves to a parked page before concluding “junk”.
8. **Self-learning write-back (with public-domain bypass)**. The parsed verdict is appended to the matching tabs — a “new domains” log (with the AI reasoning) and a “trusted domains” list — so a domain that needed an AI call today becomes a free internal hit tomorrow. A per-branch `Bypass Sheet` gate handles the special case: when the identified company came from a **public domain**, the workflow skips the trusted-domain lists (a shared webmail domain must never be cached as a company's domain) and instead records it in a dedicated **Public Domains Classified** tab. Junk verdicts are appended to the junk list.
9. **Merge (4 inputs)**. All four outcome paths (junk, internal, AI carrier/shipper, AI junk) are consolidated through a single append-mode Merge node, giving every downstream node one reliable line of ancestry.
10. **Closed-loop Marketo update**. The workflow fetches a fresh OAuth2 access token, then issues a REST `createOrUpdate` against the lead, writing native boolean flags and a multi-line AI summary back onto the record.

Field mapping written back to Marketo

Source (\$('Merge'))	Target field (placeholder)	Payload formatting
is_carrier	is_carrier_contact	Native boolean (true / false)
is_shipper	is_shipper_contact	Native boolean (true / false)
classification_status + ai_reasoning	aiQualificationSummary	Multi-line string via JSON.stringify()

Field names shown are neutral placeholders; the live integration uses the organisation's own custom field API names.

Reference data (Google Sheets)

The classifier is backed by a shared Google Sheet whose tabs act as both a reviewed allow-list and a self-growing cache. Previously the team **maintained the two trusted-domain tabs by hand** — every known carrier or shipper domain was added manually to “carriers trusted domains in marketo” or “shippers trusted domains in marketo”. The workflow now keeps these in sync automatically and adds several derived tabs:

- **new carrier domains / new shipper domains** — every AI-classified domain, logged with its reasoning.
- **carriers / shippers trusted domains in marketo** — the reviewed allow-lists, now appended by the flow instead of by hand.
- **Public Domains Classified** — public / free-mail domains where the company was still identified, kept out of the trusted lists.
- **junk domains** — the exclusion list of unusable domains.

+	≡	📄	AR to be send ▾	declining to be sent ▾	new carrier domains ▾	new shipper domains ▾	not carrier domain ▾
▾			carriers trusted domains in marketo ▾	shippers trusted domains in marketo ▾	Public Domains Classified ▾	junk domains ▾	

The reference sheet's tabs — trusted-domain allow-lists (once manual) alongside the flow's auto-maintained tabs.

Example alert output

Once a lead is classified, Marketo raises a qualification alert carrying the flags and the full AI corporate profile. A representative alert (all data replaced with neutral placeholders):

Lead Qualification Alert

Person Data:

Name: [redacted]

Email: office@companya-transport.example

Company: COMPANY A TRANSPORT LTD

Job Title: Director

Phone:

Country: [United Kingdom](#)

AI Overview:

AI Reasoning: AI Web Search

Classification Status:

Carrier - Company A Transport Ltd is a UK-based logistics provider offering general contract haulage, full and part load transport, traction, and same-day delivery services.

Corporate Profile:

- company_name: COMPANY A TRANSPORT LTD
- industry_description: Road freight transport services, including contract haulage, same-day delivery, traction services, and cargo storage
- nace_code: 4941
- nace_description: Freight transport by road
- logistics_classification: Carrier
- entity_location: [United Kingdom](#)
- entity_annual_revenue_eur: [Unknown](#)
- ultimate_parent_name: COMPANY A TRANSPORT LTD
- ultimate_parent_global_revenue_eur: [Unknown](#)

Sources: <https://companya-transport.example/>, [<COMPANIES_REGISTRY_URL>](#)

Tools & techniques

- **Three-tier deterministic** → **AI routing** — cheap junk and internal-DB checks resolve the bulk of traffic for free; the model is reserved for the genuine long tail.
- **Junk-exclusion tier** — a first-class “junk” outcome keeps free-mail, parked and unidentifiable domains out of the carrier/shipper pipeline.
- **Public-domain handling** — free-mail domains skip the trusted-domain cache and are researched directly; if the company is still identifiable the result is filed under Public Domains Classified rather than polluting the trusted allow-lists.
- **Retrieval-augmented classification** — Gemini is grounded with Google Search and a brand-name re-search fallback, so answers reflect the real company rather than stale model memory.
- **Self-growing dataset** — dual write-back (reasoning log + trusted list) turns the reference sheet into a continuously improving asset.
- **Stable multi-branch merge** — a 4-input append-mode Merge collapses mutually exclusive paths into one ancestry line, eliminating “node not executed” reference errors.
- **Programmatic OAuth2** — a fresh token is pulled per run from Marketo's identity endpoint, side-stepping the infamous 602-inside-200 expiry problem with zero manual credential saves.
- **Native boolean mapping** — comparison expressions are mapped unquoted so n8n injects real JSON booleans into Marketo checkbox fields.
- **Provenance tracking & secret externalisation** — every response records how it was classified, and all keys live in encrypted custom-auth credentials, never in the exported JSON.

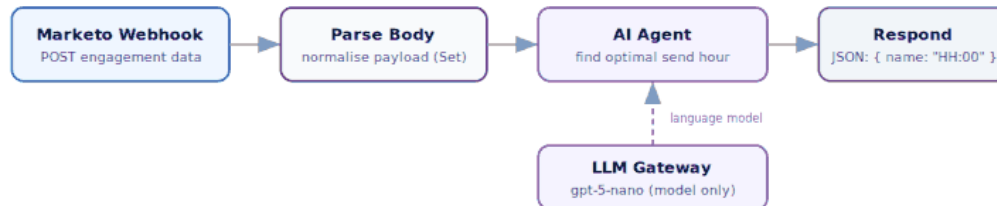
Why it's interesting

It uses an LLM **surgically**: deterministic layers handle most traffic for free, the model is constrained tightly enough to be machine-readable, and the system gets cheaper, smarter and more autonomous the longer it runs — closing the loop all the way back onto the live Marketo record.

Workflow 3 — Email Send-Time Optimization

Platform: n8n · **Trigger:** Webhook (POST, retry-on-fail) · **Pattern:** Marketo round-trip + LLM-as-calculator

Workflow 3 — Email Send-Time Optimization (n8n)



What it does

This workflow answers one narrow, high-value question per contact: **“What is the single best hour of the day to e-mail this person?”** Marketo calls the workflow with the contact's historical engagement timestamps, the workflow computes the optimal hour, and hands a single clean value straight back to Marketo to store on the lead.

Step-by-step logic

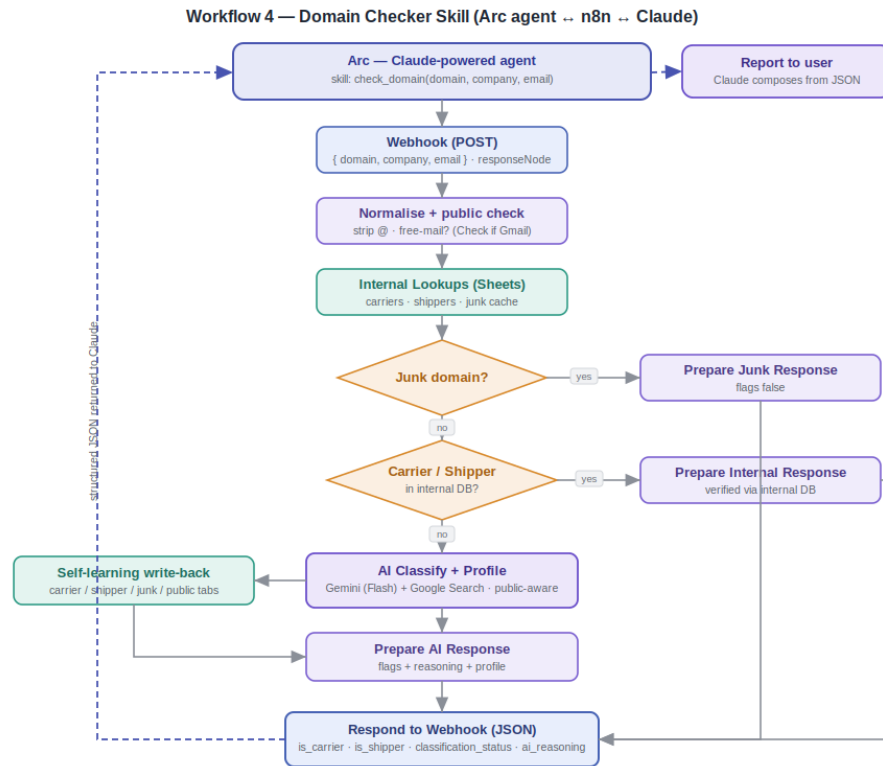
1. **Marketo webhook (POST)** receives the engagement payload (open / click timestamps); configured with retry-on-fail (up to 2 tries).
2. **Parse webhook body** — a Set node normalises the incoming body into a clean object.
3. **AI Agent** analyses the data: prioritise click timestamps over opens, find the peak engagement time, round to the nearest whole hour, and respond with exactly five characters in **HH:00** 24-hour CEST — no prose.
4. **LLM gateway** — the agent's reasoning is served through an internal model gateway, wired in as the language model only; all logic lives in the system prompt.
5. **Respond to Marketo** — returned as { "name": "HH:00" }, matching the response shape a Marketo webhook maps directly onto a lead field.

Why it's interesting

It shows restraint: rather than letting a model ramble, it turns messy engagement history into a **single, perfectly formatted value** an existing platform can act on with zero post-processing.

Workflow 4 — Domain Checker Skill (Arc agent ↔ n8n ↔ Claude)

Platform: n8n · **Trigger:** Webhook (POST, response-node) · **Pattern:** classifier exposed as an Arc skill; Claude calls it and turns the returned JSON into a report



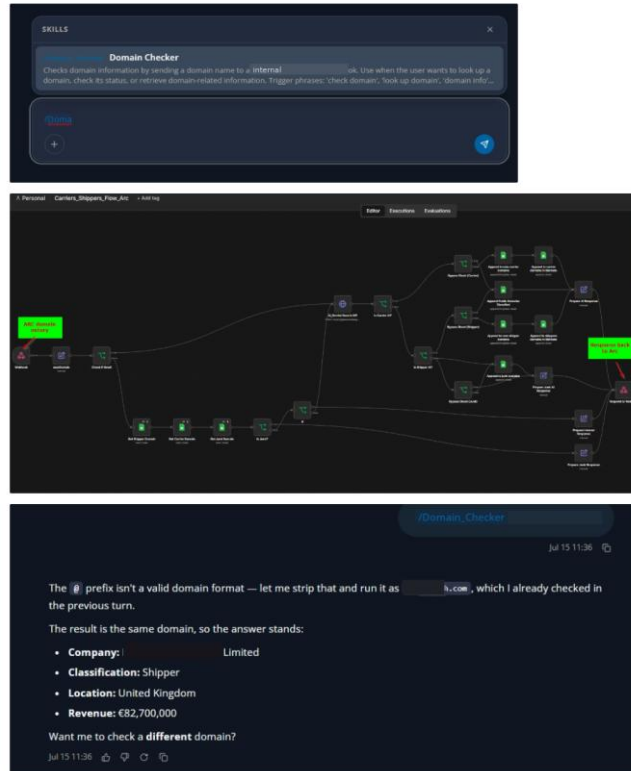
About Arc — the agent platform

Arc is the organisation's internal AI-agent platform: a single place for employees to get work done across their tools using natural language instead of switching between apps. It works through **skills** — purpose-built capabilities that connect to services such as Gmail, Slack, Jira, Google Calendar and Notion — so people can read email, manage tickets, schedule meetings, search documents and automate workflows just by asking.

Internally it is tailored to the organisation's own processes and products, with skills wired to the internal Jira instance, Marketo, n8n automations and internal product lines. Arc is **powered by Claude**, so the Domain Checker skill closes a full loop: Claude calls the n8n workflow, n8n classifies the domain and returns structured JSON, and Claude turns that JSON into a readable report for the user.

In Arc, the skill is registered as **Domain Checker** and can be invoked conversationally (for example `/Domain_Checker <domain>`) — letting a person review a single domain on demand, without pushing anything through the Marketo flow:

- Arc flow - Domain Checker skill, that allows person for manual domain review, without need to push via the Marketo flow.



Top: the skill registered in Arc. Middle: the n8n flow behind it (entry from Arc, response back to Arc). Bottom: an example conversation returning the classification and profile (domain and company redacted).

What it does

This skill exposes the three-tier classifier (carrier / shipper / junk, with public-domain handling) as a callable Arc skill. Because Arc is powered by Claude, the flow is a full round-trip: Claude calls `check_domain` with the domain (plus optional company and email), the n8n workflow classifies it and returns structured JSON synchronously, and Claude composes a readable report for the user. Where Workflow 2 pushes its verdict onto a Marketo lead, this variant hands the verdict back to the agent.

Step-by-step logic

1. **Agent tool call.** Claude (the Arc agent) POSTs { domain, company, email } to the webhook (response-node mode) over OAuth2 client-credentials, with a generous timeout to cover the research step.
2. **Normalise domain (Set).** Strips any @ prefix or protocol and lower-cases the value.
3. **Internal lookups (Google Sheets).** The carrier, shipper and junk tabs act as a cache for fast repeat lookups.
4. **Junk / internal checks.** A junk hit or an internal carrier/shipper hit short-circuits to a prepared response — no AI cost.
5. **AI classify + profile (fallback).** Unknown domains are researched by Gemini with Google Search grounding and the same strict Carrier / Shipper / Junk output contract and brand-name fallback used in Workflow 2.
6. **Self-learning write-back.** New verdicts are appended to the reference tabs so future calls resolve from cache.

7. **Respond to webhook.** A single JSON object — `is_carrier`, `is_shipper`, `classification_status` and `ai_reasoning` (the full corporate profile) — is returned to the agent.
8. **Report generation (Claude).** Arc receives the JSON and Claude composes a concise, readable report — classification, reasoning and the corporate profile — for the user in the same conversation.

Response contract

Field	Meaning
<code>is_carrier</code>	"true" / "false" — domain provides transport / freight-forwarding services
<code>is_shipper</code>	"true" / "false" — domain makes or sells goods that need moving
<code>classification_status</code>	Carrier / Shipper / Junk headline plus short reasoning and provenance
<code>ai_reasoning</code>	Structured profile: company_name, industry, NACE code/description, location, revenue (EUR), ultimate parent, sources

Tools & techniques

- **Skill-as-a-service** — the workflow is consumed as one clean agent tool; all branching complexity is hidden behind a single request/response.
- **Synchronous round-trip** — response-node mode returns the verdict in the same call, so the agent can act on it without polling or a callback.
- **Shared classifier core** — the junk/internal/AI tiers and self-learning write-back are reused verbatim from Workflow 2, so both the Marketo push and the agent pull stay in lock-step.
- **Cache-first economics** — the internal tabs double as a lookup cache, so repeat domains return instantly and cheaply.
- **Resilient by design** — a generous timeout absorbs the occasional slow research call on a fresh, uncached domain.
- **Credential externalisation** — the agent authenticates with OAuth2 client credentials; no secrets live in the workflow export.
- **Closed loop with Claude** — the agent both calls the workflow and turns its JSON into a written report, so a back-office pipeline becomes a conversational answer.

Operational notes

- The workflow must be active and published for the webhook to answer; a deactivated flow returns 404, and unpublished edits are not picked up.
- Batch lookups are not supported — one domain per call.
- Revenue figures are AI-estimated and can vary slightly between runs, so they are indicative rather than an authoritative source for financial due diligence.

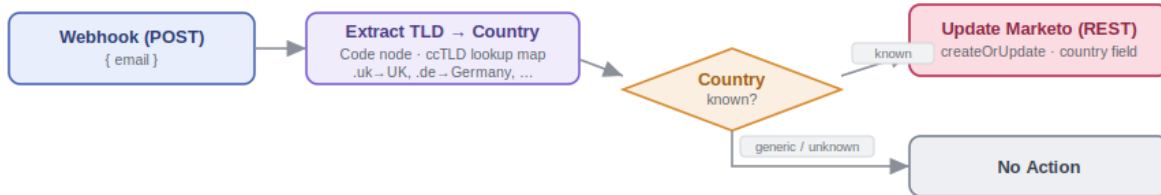
Why it's interesting

It turns a back-office data pipeline into a **first-class agent capability**: the exact same classifier that enriches Marketo records is now something a human can ask for conversationally, with the deterministic cache keeping the interactive experience fast and cheap.

Workflow 5 — Email TLD → Country Updater

Platform: n8n · **Trigger:** Webhook (POST) · **Pattern:** lightweight deterministic enrichment

Workflow 5 — Email TLD → Country Updater (n8n)



What it does

A small, dependency-free enrichment: given a lead's e-mail, it derives a likely country from the address's country-code top-level domain (ccTLD) and writes it back to Marketo. It is the kind of cheap, deterministic helper that fills obvious gaps without an AI call.

Step-by-step logic

1. **Webhook (POST)** receives `{ "email": "..." }`.
2. **Extract TLD & map country (Code)**. A Code node takes the part after the final dot, lower-cases it, and maps common ccTLDs (uk → United Kingdom, de → Germany, fr → France, and so on) to a country; anything unmapped (.com, .net, ...) becomes "Generic/Unknown".
3. **Is country known?** A guard splits on the result.
4. **Update Marketo (REST) / No Action**. A recognised country is written to the lead's country field via a `createOrUpdate` call (looked up by e-mail); generic/unknown TLDs take a no-op branch so the field is never overwritten with a guess.

Tools & techniques

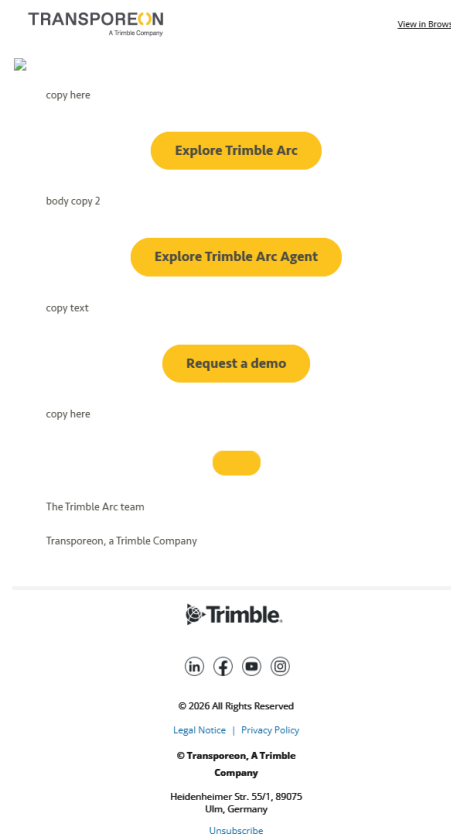
- **Deterministic-first** — a plain lookup table resolves the common cases instantly and for free; no model is involved.
- **Safe defaults** — ambiguous TLDs are routed to a no-op rather than writing a low-confidence value, so the flow never degrades existing data.
- **Same write-back contract** — reuses the Marketo REST `createOrUpdate` pattern (lookup by e-mail) shared across the portfolio.

Why it's interesting

It is a reminder that not every enrichment needs AI: a tiny, auditable rule table plus a guarded write-back covers a whole class of leads cheaply and predictably, and it composes with the heavier classifier workflows on the same Marketo instance.

Skill — Marketo Email Builder (Word brief → Marketo)

Type: Custom Claude / Cowork skill · **Target:** Marketo (via MCP) · **Input:** Word (.docx) brief



The master email template the skill targets. From a Word brief the skill fills the banner, body-copy and CTA slots and assembles the finished, on-brand email.

What it does

A custom skill I authored to turn a plain Word brief into a Marketo-ready email built against the organisation's master template. Instead of hand-copying text into Marketo's editor, the author writes the email in a **.docx** — headline, body sections, and calls-to-action annotated as **[cta_1]**, **[cta_2]**, ... — and the skill maps each piece onto the correct template slot, keeping the layout, footer and styling consistent every time.

It handles single- and multi-language campaigns (for example EN, DE, ES, FR, IT, PL) from one brief, and covers common shapes like launch announcements, nurture sequences and demo confirmations.

Two modes

- **Mode A — HTML snippets.** Produces paste-ready HTML blocks for manual entry via Marketo's Edit Code, when someone prefers to assemble the email by hand.
- **Mode B — end-to-end via Marketo MCP.** Clones the program and email, sets program tokens, uploads the banner, routes each CTA to its slot, deletes unused slots, and swaps in the right footer snippet — producing the finished Marketo asset directly.

Example

A representative use (illustrative, with placeholder content): a three-CTA product email built from a one-page Word brief. The author writes the copy with [cta_1], [cta_2] and [cta_3] annotations; the skill maps them onto the template's three CTA slots, removes the unused fourth slot, uploads the banner, sets the program tokens, and generates an EN and a DE variant from the same brief — ready to review in Marketo.

This skill is referenced here to document the capability; no email was generated for this portfolio.

Tools & techniques

- **Brief-as-source-of-truth** — one .docx drives the copy, CTAs and language variants, so content lives in a reviewable document rather than in the email editor.
- **Slot routing via annotations** — [cta_N] tags map copy to fixed template slots, and unused slots are pruned automatically.
- **Master-template fidelity** — every email inherits the same header, footer and styling, so output stays consistent and on-brand.
- **MCP automation** — Mode B drives Marketo programmatically (clone, tokens, banner upload, footer-snippet swap) with no manual copy-paste.
- **Documented platform workarounds** — absorbs Marketo quirks such as the subject-line length limit, a token UTF-8 encoding bug, the nested-token limitation and a module-variable API gap.

Why it's interesting

It compresses a fiddly, error-prone manual process — rebuilding the same templated email over and over, in several languages — into a single Word brief, while keeping the output pixel-consistent with the master template.

Backfilling 10k domains at scale (Claude → Arc → n8n → Arc → Claude)

The same skill scales into a batch backfill. Around **10,000 domains** already in Marketo were still unclassified as carrier or shipper. Claude drives the loop end-to-end: it reads the backlog, asks the Arc Domain Checker skill for each domain, which starts the n8n flow, returns the verdict to Arc, and Claude writes the result back and regenerates a running report.

The run is **checkpointed in batches of ~150** so it can pause and resume safely, tracked in a status file with processed / to-re-run / failed / pending counts. Transient timeouts are queued for automatic retry rather than treated as failures, and progress is reported after each batch (for example, roughly a third of the list processed with no quota trips on the dedicated key).

▼ Current Status



Identified 10k domains in Marketo not yet classified as Shipper or Carrier.

Using flow that connects from Claude to Arc - Ask Domain Checker - to Start n8n flow and push back the report:

Where you are

- ~29% done (3,139 / 10,829)
- ~1,100 domains processed today, still no quota trips on the dedicated key

All checkpointed. Want another 150, or wrap for now?

Background shell completed Launch another 150-domain run

Read outcome and regenerated report >

Done ☒

Status — [processing_status.csv](#)

Status	Count
☒ Processed	3,284
☐ To re-run	5
☒ Failed	0
☐ Pending	7,540

145 finished + 5 TO_RERUN in 2h 40m. The 5 are scattered transient timeouts (no streak, circuit breaker didn't trip) — they'll auto-clear first on the next run.

Where you are

- ~30% done (3,284 / 10,829) — nearly a third of the list
- ~1,240 domains today, still no quota trips on the dedicated key

All checkpointed. Want another 150, or wrap here?

🔍 ⌕ ⌕ 1.6 hours ago

let's push another 150 domains

File

processing_status.csv

domain	status	is_carrier	is_shipper
acme-freight.example	processed	false	true
nord-logistics.example	processed	false	true
via-cargo.example	processed	false	true
fjord-shipping.example	processed	false	true
fontana-foods.example	processed	false	true

Progress tracking for the batch backfill: a live status table (processed / to-re-run / failed / pending) and the per-domain results file (domains redacted).

Why it matters: it turns an interactive skill into an unattended data-cleanup pipeline — the same classifier that enriches one lead can burn down a five-figure backlog, with Claude orchestrating the loop and keeping an auditable, resumable record throughout.

Latest Achievement — Production Hardening of the Qualification Agent

The Carrier / Shipper / Junk agent was taken from a sandbox prototype to a bulletproof, production-ready enterprise integration. The work resolved five classes of failure and locked in the architectural principles below.

1. Resolved fatal JSON validation & control-character crashes

Passing Gemini's multi-line structured profile inside standard JSON double quotes crashed n8n's parser with a "Bad control character in string literal" error. The fix was the **unquoted JSON.stringify(...) pattern**: wrapping the combined text block in `JSON.stringify` and removing the outer quotes lets n8n compile the JavaScript string first and emit a perfectly escaped JSON string literal.

2. Designed a stable multi-branch routing (Merge pattern)

The workflow splits into four mutually exclusive evaluation branches, but referencing parent nodes of skipped branches threw "Node not executed" errors. A central **4-input Merge node in Append mode** placed just before the auth steps consolidates all active streams into one line of ancestry, so `${('Merge').item.json}` is stable no matter which path fired.

3. Automated OAuth2 token retrieval (immunity to error 602)

Marketo access tokens expire after 60 minutes and Marketo notoriously returns expiry (602) errors inside 200 OK payloads, bypassing background refresh. A **programmatic 2-step token flow** now pulls a fresh token from the identity endpoint immediately before each write, guaranteeing uptime with zero manual credential saves.

4. Secured keys with predefined custom auth

Rather than hard-coding the client id and secret into the HTTP node (which would leak on export), credentials live in an **encrypted generic custom-auth credential** inside n8n; the node references them through an encrypted key wrapper, keeping secrets out of the workflow JSON entirely.

5. Native boolean mappings for Marketo checkboxes

Marketo checkbox fields expect a native JSON boolean rather than the strings `"true"` / `"false"`. Mapping the comparison expression unquoted (e.g. `"is_carrier_contact"`: `{{ $('Merge').item.json.is_carrier === 'true' }}`) makes n8n evaluate it at runtime and inject a raw, unquoted boolean.

Outcome

The agent now classifies into three tiers, teaches itself by writing every new verdict back into the reference dataset, refreshes its own Marketo credentials, and updates the live lead record synchronously — running unattended in production.